

Course: B.Tech.

Semester: IV

Branch: EE/MME/ME

Subject: Mathematics-IV

Subject Code: MTH 241

Max. Marks: 50

Max Time: 180 Minutes

Note: Attempt all the questions. Usual notations are used.

- 1(a). i). Find all basic solutions and identify the basic feasible solutions (BFS) and degenerate basic feasible solutions of the following system of equations:

$$2x + y + 4z = 11$$

$$3x + y + 5z = 14$$

- ii). Use simplex method to solve the following LPP:

$$\text{Maximize } Z = 4x_1 + x_2$$

Subject to the constraints:

$$x_1 - x_2 \leq 1;$$

$$-2x_1 + x_2 \leq 2; \text{ and } x_1, x_2 \geq 0.$$

- (b). A small furniture workshop manufactures three items: Tables (x_1 units), Chairs (x_2 units) and Stools (x_3 units). The cost to produce each unit of Tables, Chairs, and Stools are 7\$, 8\$, and 3\$, respectively. The workshop wants to minimize the total production cost while meeting specific order requirements:

- i) A showroom display requires at least 2 items made up of Tables and Chairs combined

$$\text{i.e. } x_1 + x_2 \geq 2.$$

- ii) A separate custom order requires that the combination of twice the number of Tables plus the number of Stool must be at least 5 units

$$\text{i.e. } 2x_1 + x_3 \geq 5.$$

How many Tables, Chairs, and Stools should the workshop produce to minimize the total production cost while satisfying the showroom and custom order requirements?

- 2(a). Solve the following linear programming problem (LPP) by dual simplex method to find the optimal solution:

$$\text{Minimize } Z = 6x_1 + 7x_2 + 3x_3 + 5x_4$$

Subject to the constraints:

$$5x_1 + 6x_2 - 3x_3 + 4x_4 \geq 12$$

$$x_2 + 5x_3 - 6x_4 \geq 10$$

$$2x_1 + 5x_2 + x_3 + x_4 \geq 8$$

$$x_1, x_2, x_3, x_4 \geq 0$$

- (b). Solve the following Integer programming problem using Branch and Bound method:

$$\text{Maximize } Z = 5x_1 + 6x_2$$

Subject to the constraints:

$$x_1 + x_2 \leq 5$$

$$4x_1 + 7x_2 \leq 28$$

$$x_1, x_2 \text{ are non-negative integers.}$$

- 3(a). JoShop wants to allocate four different categories of machines to five types of tasks. The numbers of machines available in the four categories are 25, 30, 20, and 30. The machines required for different jobs in the five tasks are 30, 10, 20, 25, and 20. The below table provides the unit cost (in dollars) of assigning a machine category to a task type.

The objective of the problem is to determine the optimum number of machines in each category to be allocated to each task type to minimize the overall cost. Solve the problem and interpret the solution.

		Task Type				
		1	2	3	4	5
Machines	A	10	2	3	15	9
	B	5	10	15	2	4
	C	15	5	14	7	15
	D	20	15	13	20	8

- (b). A fast-food chain wants to open four stores in six cities. The expected profits from each store (in lakhs) is as shown in the following table:

	Cities					
	I	II	III	IV	V	VI
A	85.3	88.0	87.5	82.4	89.1	86.7
B	78.7	77.4	77.4	76.5	79.3	78.3
C	82.0	81.3	82.4	80.6	83.5	81.7
D	84.3	84.6	86.2	83.3	84.4	85.5

How the stores should be allocated to the cities to maximize the profit.

- 4(a). The payoff matrix of a game played between Player A and Player B is given in the following table. Find the best strategies of each player and value of the game by solving the game using graphical method:

		Player B	
		B_1	B_2
Player A	A_1	1	-3
	A_2	3	5
	A_3	-1	6
	A_4	4	1
	A_5	2	2
	A_6	-5	0

- (b). Solve the following game between Player A and Player B using the principle of dominance, whose payoff matrix is given below. Include in your answer:

- Does the game have any pure strategy or saddle point?
- Optimal strategy selection for each player.
- The value of the game.
- Is it a fair game? If not discuss the observed bias.

		Player B					
		B_1	B_2	B_3	B_4	B_5	B_6
Player A	A_1	4	2	0	2	1	1
	A_2	4	3	1	3	3	3
	A_3	4	3	7	-5	1	2
	A_4	4	3	4	-6	2	2
	A_5	4	3	3	-8	2	2

- 5(a). Solve the following linear programming problem

$$\text{Maximize } Z = 4x_1 + 5x_2$$

Subject to the constraints:

$$4x_1 + 5x_2 \leq 20$$

$$7x_1 + 3x_2 \leq 21$$

$$x_1, x_2 \geq 0$$

Also, apply the concept of sensitivity analysis to find the optimal solution of the given LPP if

- Right hand side vector $[20, 21]^T$ is changed to $[1, 2]^T$.
- Right hand side vector $[20, 21]^T$ is changed to $[5, 1]^T$.

- (b). Solve the following non-linear programming problem

$$\text{Optimize } Z = -x_1^2 - x_2^2 - x_3^2 + 12x_1 + 8x_2 + 6x_3 - 23$$

Subject to the constraints:

$$x_1 + x_2 + x_3 = 10$$

$$x_1, x_2, x_3 \geq 0$$

*****End of The Paper*****